

ExempleCallable (v2)

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Objectif de cette version...

2

Variante

-  Insérer des délais
-  Attendre le résultat un «certain temps»
-  Juste pour voir



toujours
facile!



La classe calcule

3

```
import java.util.Random;
import java.util.concurrent.Callable;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;
import java.util.concurrent.Future;
import java.util.concurrent.TimeUnit;
import java.util.concurrent.TimeoutException;

class Calcule implements Callable<Integer> {
    String monNom;
    private Random generateur = new Random();

    Calcule (String n) {
        monNom = n;
    }

    public Integer call() {
        Integer resultat = generateur.nextInt(1000);
        System.out.println(monNom + " rend " + resultat);
        try { // Petite attente (pour tester les time-out)
            Thread.sleep(resultat);
        }
        catch (Exception e) {
            System.out.println("Ne devrait pas arriver " + e);
        }
        return resultat;
    }
}
```

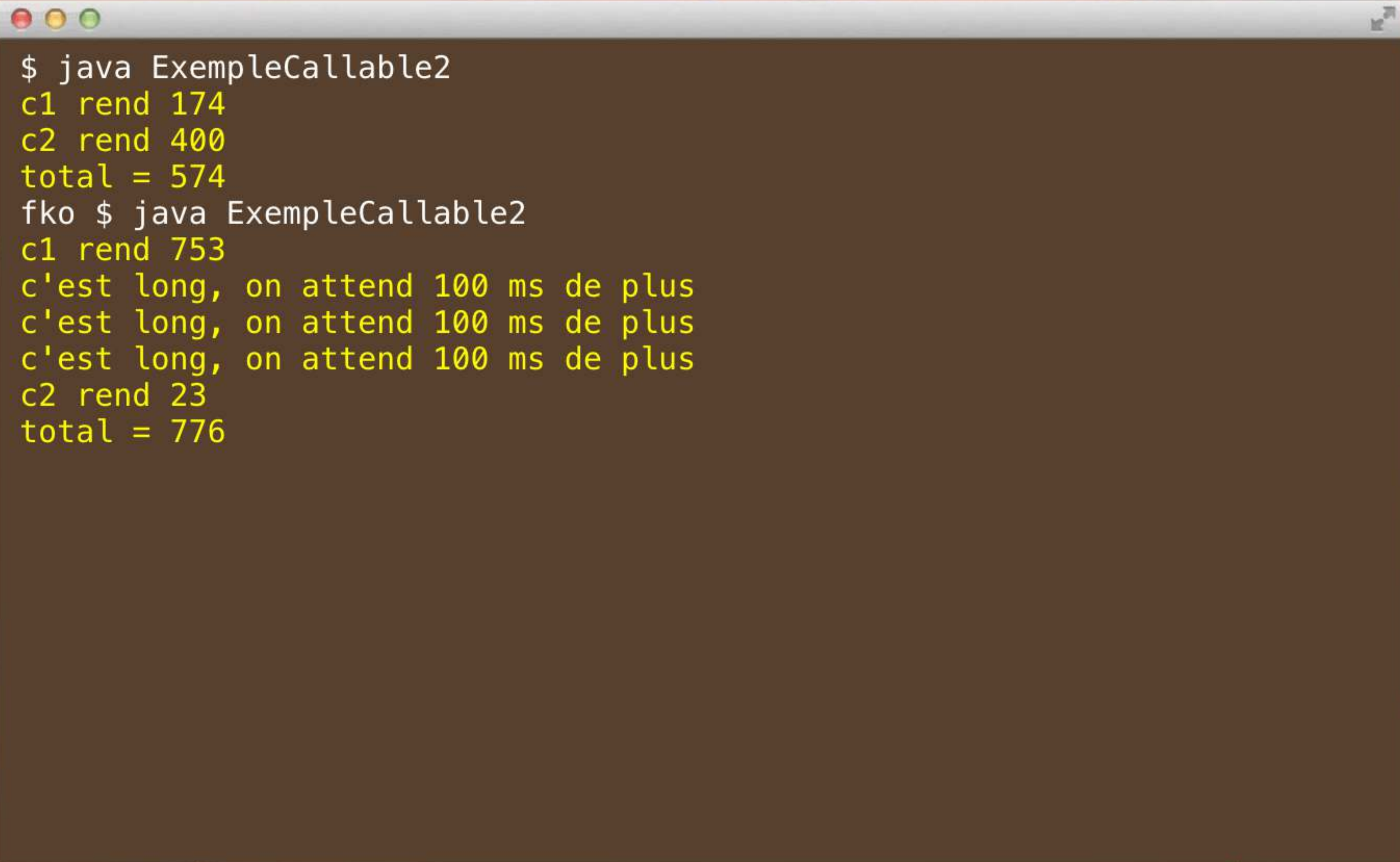
La «classe principale»

4

```
public class ExempleCallable2 {
    public static void main(String []args) {
        Integer total = 0;
        ExecutorService executor = Executors.newFixedThreadPool(3);
        try {
            Future<Integer> f1 = executor.submit(new Calcule("c1"));
            int duree = 500;
            boolean reprendre = true;
            while (reprendre) { // On boucle sur de l'impatience ;- )
                try {
                    total = total + f1.get(duree, TimeUnit.MILLISECONDS);
                    reprendre = false;
                }
                catch (TimeoutException e) {
                    duree = 100;
                    System.out.println("c'est long, on attend 100 ms de plus");
                }
            }
            // 2ème appel normal
            Future<Integer> f2 = executor.submit(new Calcule("c2"));
            total = total + f2.get();
            System.out.println("total = " + total);
            executor.shutdown();
        }
        catch (Exception e) {
            System.out.println("oulala !!! " + e);
        }
    }
}
```

Exécutions...

5

A terminal window with a dark background and light-colored text. The window has a title bar with three colored buttons (red, yellow, green) on the left and a maximize button on the right. The text inside the terminal shows the execution of a Java program named 'ExempleCallable2'. The first execution shows 'c1' ending at 174 and 'c2' ending at 400, with a total of 574. The second execution, preceded by 'fko', shows 'c1' ending at 753 and 'c2' ending at 23, with a total of 776. Between the two executions, there are three lines of text: 'c'est long, on attend 100 ms de plus'.

```
$ java ExempleCallable2
c1 rend 174
c2 rend 400
total = 574
fko $ java ExempleCallable2
c1 rend 753
c'est long, on attend 100 ms de plus
c'est long, on attend 100 ms de plus
c'est long, on attend 100 ms de plus
c2 rend 23
total = 776
```